Assisted Grading Architecture

System Specification for Local, DSGVO-Compliant Exam Processing

1. System Overview

This document outlines the end-to-end technical architecture for an automated, AI-assisted grading pipeline (*Klassenarbeitskorrektur*) designed for local school server deployments. Engineered specifically for rigorous subjects like mathematics and physics, the system processes handwritten student exams overnight. By morning, it presents educators with a structured, pre-evaluated digital copy that requires only rapid manual verification.

2. The Data Ingestion & Pre-Processing Pipeline

The pipeline begins immediately after scanned PDFs of the student exams are uploaded to the local server. Processing occurs during off-peak night hours to maximize the utilization of existing educational IT infrastructure.

- **Geometric Deskew & Line Removal:** OpenCV identifies the dominant angle of the ruled paper and rotates the image to a perfect horizontal baseline. An in-painting algorithm removes the pre-printed ruled lines, preventing interference with character descenders.
- **Layout Segmentation:** A localized Vision Model separates paragraphs of natural language text from spatial mathematical equations and hand-drawn diagrams.
- **Math OCR & LaTeX Extraction:** Mathematical segments are processed through an engine (such as GOT-OCR 2.0). Raw ink is converted directly into structured *LaTeX* strings.
- **Vectorization Isolation:** Free-body diagrams and graphs are isolated as localized image snippets to be evaluated alongside the text.

3. The Tripartite Context Engine (Primary Evaluation)

A raw transcription is insufficient for AI grading. The core breakthrough of this system lies in the contextual ingestion phase. The primary Large Language Model (LLM) is provided with three distinct inputs simultaneously:

Input A: The Clean Digital Copy. The output from the OCR pipeline—a highly structured Markdown document containing text and LaTeX equations, minimizing hallucination risks from raw handwriting.

Input B: The Unsolved Question Paper. The original, blank exam document. This provides the AI with the exact phrasing, constraints, and point distributions the student was given.

Input C: The Guide Solution (Erwartungshorizont). The official rubric and model answers provided by the teacher, dictating exact criteria for partial and full credit.

Using this tripartite context, the Primary AI assesses the student's logic step-by-step, generating localized feedback, identifying exact points of mathematical failure, and proposing a preliminary score.

4. Dual-Verification Architecture

To eliminate the risk of "AI hallucinations" unfairly docking student points, the system employs a secondary, isolated verification step.

- **Syntactic Validation:** Before AI evaluation, libraries like `SymPy` verify the mathematical syntax of the generated LaTeX. If an equation is mathematically malformed (e.g., missing brackets due to bad OCR), it is flagged instantly.
- **The Auditor AI:** A second, independently prompted LLM acts as an auditor. It receives the Primary AI's proposed feedback and score, and cross-references them strictly against the Guide Solution. If the Auditor detects that the Primary AI missed a valid alternative solution path, or incorrectly applied the rubric, the system highlights the discrepancy for human review.

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[STUDENT EXAM] + [BLANK PAPER] + [RUBRIC]
      |                 |              |
      ▼                 ▼              ▼
┌──────────────┐  ┌──────────────────────────┐
│ OCR Pipeline ├─►│  Primary AI Evaluator    │ │
└──────────────┘  └──────────────────────────┘
                         │ (Proposed Score & Feedback)
                         ▼
                  ┌──────────────────────────┐
                  │ Auditor AI Verification  │ (Checks against Rubric)
                  └──────────────────────────┘
                         │
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                  ┌──────────────────────────┐
                  │ Morning Dashboard UI     │ (Teacher Approval)
                  └──────────────────────────┘
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5. The Morning Review (Human-in-the-Loop)

By morning, the local server has completed batch processing. The teacher logs into a web-based dashboard that strictly enforces the legal standard that final grading authority rests with a certified educator.

The dashboard highlights low-confidence OCR reads and auditing discrepancies in a dedicated triage queue. For high-confidence answers, the teacher views the original handwritten snippet side-by-side with the AI's proposed score and rubric justification. A single click of "Approve" locks in the grade. This pipeline reduces a 20-minute per-paper grading cycle to less than 4 minutes, ensuring scalable deployment across public Gymnasiums while maintaining absolute DSGVO compliance.